Concept 3 | Properties and Graphs of Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Period

$$T = \frac{2\pi}{|b|}$$

Amplitude

$$y = a\sin(bx + c) + d \Rightarrow |a|$$

Range

$$d - |a| \le y \le d + |a|$$

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Classified Practice **Properties and Graphs of Trigonometric Functions**

Q001 **Written**

(1) Express $\sin\frac{9\pi}{5}$ **as a trigonometric function of an acute angle. (2) Simplify** $\sin\theta + \sin(\theta+\frac{\pi}{2}) + \sin(\theta+\pi) + \sin(\theta+\frac{3\pi}{2})$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q002 **Written**

(1) Express $\cos(-\frac{\pi}{6})$ **as a trigonometric function of an acute angle. (2) Express** $\sin\frac{8\pi}{7}$ **as a trigonometric function of an acute angle. (3) Simplify** $\cos(\frac{\pi}{2}-\theta)\sin(\pi-\theta)-\sin(\frac{3\pi}{2}+\theta)\cos(\pi-\theta)$**.**

Classified Practice Properties and Graphs of Trigonometric Functions

Q003 Written

Solve all three given demands: (1) express $\cos\frac{17\pi}{5}, \cos\frac{5\pi}{7}, \sin(-\frac{7\pi}{8}), \sin(-\frac{19\pi}{8})$ **using acute angles; (2) find** $\tan(-\frac{\pi}{6})$ **and** $\tan(-\frac{\pi}{4})$; **(3) simplify (a)** $\sin\theta + \cos(\pi+\theta) - \cos(\pi-\theta) + \sin(\pi+\theta)$, **(b)** $\sin(\theta+\frac{\pi}{2})\cos(\theta+\pi) + \sin(-\theta)\cos(\frac{\pi}{2}-\theta)$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q004 **Written**

Simplify $\sin(\frac{3\pi}{2} - \theta)\sec(\pi + \theta) - \tan(\theta - \frac{\pi}{2})\cot(\frac{5\pi}{2} + \theta)$.

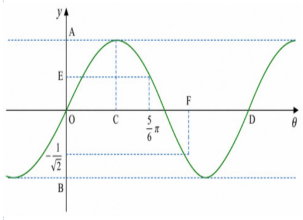
Q005 **Written**

Draw the parent graphs $y = \sin\theta$, $y = \cos\theta$, and $y = \tan\theta$, and state each period and range.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q006 **Written**

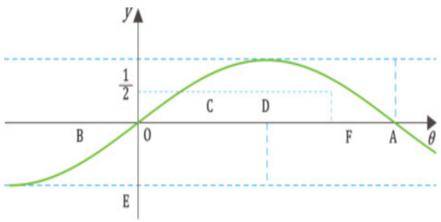
For the given graph of $y = \sin\theta$, determine the labelled scales A to F.



Classified Practice Properties and Graphs of Trigonometric Functions

Q007 Written

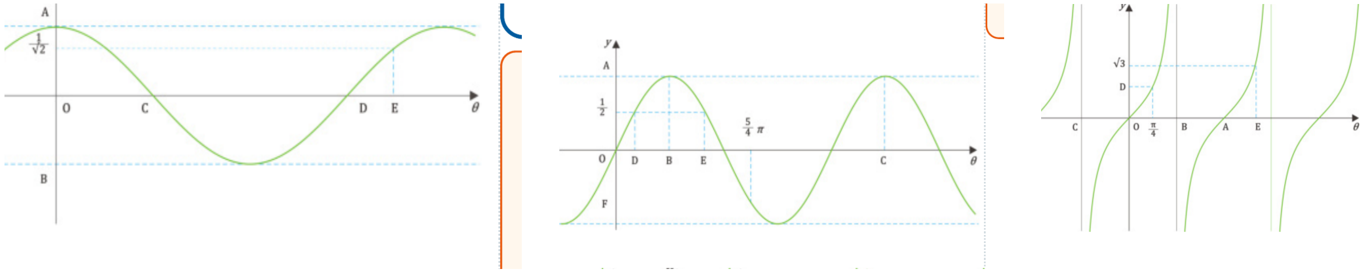
(1) Draw the parent graphs $y = \sin\theta$, $y = \cos\theta$, $y = \tan\theta$, and state their periods. (2) For the given graph of $y = \cos\theta$, determine the labelled scales A to F.



Classified Practice **Properties and Graphs of Trigonometric Functions**

Q008 Written

Complete all four given graph tasks: draw the three parent graphs and find their periods; read the labelled scales on the given cosine graph; read the labelled scales on the given sine graph; and read the labelled scales on the given tangent graph.

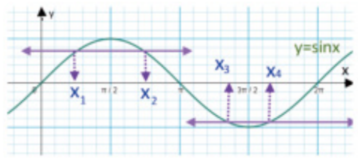


Classified Practice **Properties and Graphs of Trigonometric Functions**

Q009 **Written**

The given graph represents $y = \sin x$ for $0 < x < 2\pi$. Find $x_1 + x_2 + x_3 + x_4$ from the four marked x-coordinates.

$y = sinx,\ 0 < x < 2\pi$



Then find: $x_1 + x_2 + x_3 + x_4$

Classified Practice   **Properties and Graphs of Trigonometric Functions**

Q010 **Written**

Draw the graphs of $y = 2\cos\theta$ and $y = \tan\frac{\theta}{2}$, and find their periods.

Q011 **Written**

Draw the graphs of $y = 2\sin\theta$, $y = \tan 2\theta$, and $y = \cos\frac{\theta}{2}$, and find their periods.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q012 **Written**

Draw the nine given graphs and find each period: **(a)** $3\sin\theta$, **(b)** $3\cos\theta$, **(c)** $\frac{1}{2}\cos\theta$, **(d)** $\sin 3\theta$, **(e)** $\cos 2\theta$, **(f)** $\tan 3\theta$, **(g)** $\sin\frac{\theta}{2}$, **(h)** $\cos\frac{\theta}{3}$, **(i)** $\tan\frac{\theta}{3}$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q013 **Written**

Draw $y = -2|\sin(3x)|$ and find its period.

Q014 **Written**

Draw $y = \sin(\theta - \frac{\pi}{6})$ and find its period.

Classified Practice Properties and Graphs of Trigonometric Functions

Q015 Written

Draw the graphs and find their periods: (1) $y = \sin(\theta - \frac{\pi}{3})$, **(2)** $y = \cos(\theta + \frac{\pi}{4})$, **(3)** $y = \tan(\theta - \frac{\pi}{2})$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q016 Written

Draw all given graphs and find their periods: sine group $y = \sin(\theta + \frac{\pi}{4}), \sin(\theta - \frac{\pi}{2}), \sin(\theta + \frac{\pi}{3})$**; cosine group** $y = \cos(\theta - \frac{\pi}{3}), \cos(\theta - \frac{\pi}{4}), \cos(\theta + \frac{\pi}{6})$**; tangent group** $y = \tan(\theta + \frac{\pi}{6}), \tan(\theta - \frac{\pi}{3}), \tan(\theta - \frac{\pi}{4})$**.**

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q017 **Written**

Draw $y = 3\cos|x + \frac{\pi}{3}|$ and find its period.

Q018 **Written**

Draw the graphs and find their periods: (1) $y = 2\cos(2\theta - \frac{\pi}{2})$, **(2)** $y = \tan(\frac{\theta}{2} - \frac{\pi}{6})$.

Classified Practice Properties and Graphs of Trigonometric Functions

Q019 Written

Draw the graphs and find their periods: (a) $y = 3\sin\frac{\theta}{2}$, **(b)** $y = \tan(2\theta - \frac{\pi}{2})$, **(c)** $y = 2\cos(2\theta - \frac{\pi}{3})$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q020 Written

Draw all nine given graphs and find their periods: (a) $2\cos\frac{\theta}{2}$, **(b)** $3\sin 2\theta$, **(c)** $2\cos 2\theta$, **(d)** $2\cos(\theta-\frac{\pi}{4})$, **(e)** $2\sin(\theta-\frac{\pi}{3})$, **(f)** $\tan(2\theta-\frac{\pi}{3})$, **(g)** $3\cos(2\theta-\frac{\pi}{2})$, **(h)** $3\sin(2\theta-\frac{2\pi}{3})$, **(i)** $2\cos(\frac{\theta}{2}-\frac{\pi}{4})$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q021 **Written**

Draw $y = 3\sin|2x - \pi|$ and find its period.

Q022 **MCQ**

$\sin(\theta + \pi)$ **equals:**

Choose the correct option.

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q023 MCQ

$\cos(\theta + \pi)$ **equals:**

Choose the correct option.

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

Q024 **MCQ**

$\sin(\frac{\pi}{2} - \theta)$ **equals:**

Choose the correct option.

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q025 **MCQ**

$\cos(-\theta)$ **equals:**

Choose the correct option.

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

Q026 **MCQ**

$\sin(-\theta)$ **equals:**

Choose the correct option.

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q027 **MCQ**

$\cos(\pi - \theta)$ **equals:**

Choose the correct option.

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

Q028 MCQ

$\sin(\frac{3\pi}{2} + \theta)$ **equals:**

Choose the correct option.

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q029 MCQ

$\tan(\theta - \frac{\pi}{2})$ **equals:**

Choose the correct option.

A $\tan\theta$

B $-\tan\theta$

C $\cot\theta$

D $-\cot\theta$

Q030 MCQ

The period of $y = \sin\theta$ is:

Choose the correct option.

A $\pi/2$

B π

C 2π

D 4π

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q031 **MCQ**

The range of $y = \cos\theta$ is:

Choose the correct option.

A $y \geq 0$

B $-1 \leq y \leq 1$

C $y \in \mathbb{R}$

D y>1

Q032 **MCQ**

The period of $y = \tan\theta$ is:

Choose the correct option.

A $\pi/2$

B π

C 2π

D 4π

Classified Practice Properties and Graphs of Trigonometric Functions

Q033 MCQ

A vertical asymptote of $y = \tan\theta$ occurs at:

Choose the correct option.

A $\theta = k\pi$

B $\theta = \frac{\pi}{2} + k\pi$

C $\theta = 2k\pi$

D $\theta = \frac{\pi}{4} + k\pi$

Q034 MCQ

The maximum value of $\sin\theta$ is:

Choose the correct option.

A -1

B 0

C 1

D π

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q035 **MCQ**

The x-coordinate of the first positive sine maximum is:

Choose the correct option.

A $\pi/4$

B $\pi/2$

C π

D $3\pi/2$

Q036 **MCQ**

At $\theta = 0$, $\cos\theta$ equals:

Choose the correct option.

A -1

B 0

C 1

D undefined

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q037 **MCQ**

At $\theta = 0$, $\tan\theta$ equals:

Choose the correct option.

A -1

B 0

C 1

D undefined

Q038 **MCQ**

The range of $y = 3\sin\theta$ is:

Choose the correct option.

A [-1,1]

B [-3,3]

C [0,3]

D $\mathbb{R}$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q039 MCQ

The period of $y = \sin 4\theta$ is:

Choose the correct option.

A $\pi/4$

B $\pi/2$

C π

D 2π

Q040 MCQ

The period of $y = \cos(\theta/3)$ is:

Choose the correct option.

A $2\pi/3$

B 2π

C 3π

D 6π

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q041 **MCQ**

The period of $y = \tan 5\theta$ is:

Choose the correct option.

A $\pi/5$

B π

C 5π

D $2\pi/5$

Q042 **MCQ**

The range of $y = \frac{1}{2}\cos\theta$ is:

Choose the correct option.

A [-2,2]

B [-1,1]

C [-1/2,1/2]

D [0,1/2]

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q043 **MCQ**

The graph of $y = -2\sin\theta$ is the parent sine graph:

Choose the correct option.

A shifted only

B reflected and stretched vertically

C compressed horizontally only

D undefined

Q044 MCQ

The period of $y = \tan(\theta/2)$ is:

Choose the correct option.

A $\pi/2$

B π

C 2π

D 4π

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q045 **MCQ**

The period of $y = \sin(3\theta)$ is:

Choose the correct option.

A $\pi/3$

B $2\pi/3$

C 3π

D 6π

Q046 MCQ

The graph of $y = \sin(\theta - \pi/4)$ is shifted:

Choose the correct option.

A left by $\pi/4$
B right by $\pi/4$
C up by $\pi/4$
D down by $\pi/4$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q047 **MCQ**

The graph of $y = \cos(\theta + \pi/3)$ is shifted:

Choose the correct option.

A right by $\pi/3$

B left by $\pi/3$

C up by $\pi/3$

D down by $\pi/3$

Q048 **MCQ**

A horizontal shift changes the sine period from 2π to:

Choose the correct option.

A π

B 2π

C 4π

D 0

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q049 **MCQ**

A horizontal shift changes the tangent period from π to:

Choose the correct option.

A $\pi/2$

B π

C 2π

D 4π

Q050 **MCQ**

The graph of $y = \sin(\theta + \pi/2)$ equals:

Choose the correct option.

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q051 **MCQ**

The graph of $y = \cos(\theta - \pi)$ equals:

Choose the correct option.

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

Q052 MCQ

The graph of $y = \tan(\theta - \pi/2)$ has its central zero at:

Choose the correct option.

A $\theta = 0$

B $\theta = \pi/2$

C $\theta = \pi$

D $\theta = 2\pi$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q053 MCQ

Since cosine is even, $\cos|x|$ **equals:**

Choose the correct option.

A $\cos(x)$

B $-\cos(x)$

C $\sin(x)$

D $|\cos(x)|$

Q054 **MCQ**

In $y = a\sin(b\theta - q)$, the horizontal shift is:

Choose the correct option.

A q
B q/b
C b/q
D a/b

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q055 **MCQ**

The period of $y = 2\cos(3\theta - \pi)$ is:

Choose the correct option.

A $\pi/3$

B $2\pi/3$

C π

D 2π

Q056 MCQ

The range of $y = 4\sin(2\theta - 1)$ is:

Choose the correct option.

A [-1,1]

B [-2,2]

C [-4,4]

D $\mathbb{R}$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q057 **MCQ**

For $y = \tan(2\theta - \pi/3)$, the shift is:

Choose the correct option.

A left by $\pi/3$

B right by $\pi/6$

C right by $\pi/3$

D left by $\pi/6$

Q058 MCQ

The period of $y = 3\sin(\theta/2)$ is:

Choose the correct option.

A π

B 2π

C 3π

D 4π

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q059 MCQ

The correct transformation order for $y = af(b\theta - q)$ starts with:

Choose the correct option.

A vertical scale
B horizontal shift
C horizontal scale
D reflection only

Q060 MCQ

The period of $y = \tan(4\theta + q)$ is:

Choose the correct option.

A $\pi/4$

B $\pi/2$

C π

D 2π

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q061 **MCQ**

The range of $y = 3\sin|2x - \pi|$ is:

Choose the correct option.

A [-3,3]

B [0,3]

C [-1,1]

D $\mathbb{R}$

Q062 **Written**

Express $\cos \frac{11\pi}{6}$ **as a function of an acute angle.**

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q063 **Written**

Simplify $\sin(\theta + \pi) - \cos(\frac{\pi}{2} - \theta)$.

Q064 **Written**

Simplify $\cos(\theta + \pi) + \cos(\frac{\pi}{2} - \theta)$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q065 **Written**

Evaluate $\tan(-\frac{3\pi}{4})$ using an acute reference angle.

Q066 **Written**

State the period and range of $y = \cos\theta$, then list its key points on one cycle.

Classified Practice Properties and Graphs of Trigonometric Functions

Q067 Written

State the period and asymptotes of $y = \tan\theta$ on $[-\pi, \pi]$.

Q068 **Written**

For the parent sine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q069 **Written**

For the parent cosine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Q070 **Written**

Find the period and range of $y = 4\cos\theta$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q071 **Written**

Find the period and range of $y = \sin(5\theta)$.

Q072 **Written**

Find the period and range of $y = \tan(\theta/4)$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q073 **Written**

Describe the transformations from $y = \cos\theta$ to $y = -\frac{1}{2}\cos(2\theta)$.

Q074 **Written**

Find the shift and period of $y = \sin(\theta + \frac{\pi}{6})$.

Classified Practice Properties and Graphs of Trigonometric Functions

Q075 **Written**

Find the shift and period of $y = \cos(\theta - \frac{\pi}{2})$.

Q076 **Written**

Find the central zero and period of $y = \tan(\theta + \frac{\pi}{4})$**.**

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q077 **Written**

For $y = 2\sin(4\theta - \pi)$, state the shift, period and range.

Q078 **Written**

For $y = \cos(\theta/3 + \pi/6)$**, state the shift and period.**

Classified Practice **Properties and Graphs of Trigonometric Functions**

Q079 **Written**

For $y = 3\tan(2\theta + \pi)$, state the shift and period.

Q080 **Written**

For $y = 2\cos(\theta/2 - \pi/8)$, state the shift, period and range.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Quiz MCQ 1 **Concept Quiz · MCQ**

The period of $y = \sin(2\theta - \frac{\pi}{3})$ is:

Choose the correct option.

A $\pi/2$

B π

C 2π

D 4π

Quiz MCQ 2 **Concept Quiz · MCQ**

$\sin(\theta + \pi)$ **equals:**

Choose the correct option.

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

Classified Practice **Properties and Graphs of Trigonometric Functions**

Quiz MCQ 3 **Concept Quiz · MCQ**

The range of $y = 2\cos\theta$ is:

Choose the correct option.

A [-1,1]

B [-2,2]

C [0,2]

D $\mathbb{R}$

Quiz MCQ 4 **Concept Quiz · MCQ**

The graph of $y = \cos(\theta - \frac{\pi}{4})$ is shifted:

Choose the correct option.

A left by $\frac{\pi}{4}$

B right by $\frac{\pi}{4}$

C up by $\frac{\pi}{4}$

D down by $\frac{\pi}{4}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 5 **Concept Quiz · Written**

For $y = 3\sin(2\theta - \pi)$, state the horizontal shift, period and range.

Quiz WRITTEN 6 **Concept Quiz · Written**

Simplify $\sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi)$.

Classified Practice **Properties and Graphs of Trigonometric Functions**

Quiz WRITTEN 7 **Concept Quiz · Written**

Draw one cycle of $y = \tan(\theta - \frac{\pi}{3})$ and state its period and asymptotes.

Quiz WRITTEN 8 **Concept Quiz · Written**

For $y = -2|\sin(3x)|$, state its period and range and describe the reflection.

Answer key

Use the question number shown on each page.

Q001 (1) $-\sin\frac{\pi}{5}$ (2) 0

Q002 (1) $\cos\frac{\pi}{6}$ (2) $-\sin\frac{\pi}{7}$ (3) $\sin^2\theta-\cos^2\theta$

Q003 (1) $-\cos\frac{2\pi}{5}, -\cos\frac{2\pi}{7}, -\sin\frac{\pi}{8}, -\sin\frac{3\pi}{8}$ (2) $-\frac{1}{\sqrt{3}}, -1$ (3a) 0, (3b) -1

Q004 0

Q005 $\sin, \cos:\ T=2\pi,\ -1\le y\le 1;\quad \tan:\ T=\pi,\ y\in\mathbb{R}$

Q006 $A=1,\ B=-1,\ C=\frac{\pi}{2},\ D=2\pi,\ E=\frac{1}{2},\ F=\frac{5\pi}{4}$

Q007 (1) $T_{\sin}=2\pi,\ T_{\cos}=2\pi,\ T_{\tan}=\pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph

Q008 Use the standard cycles: sine/cosine have period 2π and range $[-1,1]$; tangent has period π, range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.

Q009 Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.

Q010 $y=2\cos\theta:\ T=2\pi,\ -2\le y\le 2;\quad y=\tan\frac{\theta}{2}:\ T=2\pi$

Q011 $T=2\pi,\ \frac{\pi}{2},\ 4\pi$ respectively; ranges $[-2,2]$, $\mathbb{R}$, $[-1,1]$

Q012 (a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) π (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π; ranges follow the outside coefficients

Q013 $T=\frac{\pi}{3},\quad 0\ge y\ge -2$

Q014 shift right by $\frac{\pi}{6}$, $\quad T=2\pi$

Q015 Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.

Q016 All sine/cosine periods are 2π; all tangent periods are π. The shifts are read directly from the brackets.

Q017 $y=3\cos(x+\frac{\pi}{3})$ since cosine is even; $\quad T=2\pi$

Q018 (1) $y=2\cos 2(\theta-\frac{\pi}{4}),\ T=\pi,\ -2\le y\le 2$; (2) $y=\tan\frac{1}{2}(\theta-\frac{\pi}{3}),\ T=2\pi$

Q019 (a) $T=4\pi,\ -3\le y\le 3$; (b) $T=\frac{\pi}{2}$; (c) $T=\pi,\ -2\le y\le 2$

Q020 Periods: (a) 4π; (b) π; (c) π; (d) 2π; (e) 2π; (f) $\frac{\pi}{2}$; (g) π; (h) π; (i) 4π. Ranges are set by the outside factors.

Q021 $y=3|\sin(2x-\pi)|,\quad T=\frac{\pi}{2},\quad 0\le y\le 3$

Q022 B

Q023 B

Q024 C

Q025 A

Q026 B

Q027 B

Q028 D

Q029 D

Q030 C

Q031 B

Q032 B

Q033 B

Q034 C

Q035 B

Q036 C

Q037 B

Q038 B

Q039 B

Q040 D

Q041 A

Q042 C

Q043 B

Q044 C

Q045 B

Q046 B

Q047 B

Q048 B

Q049 B

Q050 C

Q051 B

Q052 B

Q053 A

Q054 B

Q055 B

Q056 C

Q057 B

Q058 D

Q059 B

Q060 A

Q061 B

Q062 $\cos\frac{\pi}{6}$

Q063 $-\sin\theta-\sin\theta=-2\sin\theta$

Q064 $-\cos\theta+\sin\theta$

Q065 1

Q066 $T=2\pi,\ -1\le y\le 1$; $(0,1), (\frac{\pi}{2},0), (\pi,-1), (\frac{3\pi}{2},0), (2\pi,1)$

Q067 $T=\pi$; $\theta=\pm\frac{\pi}{2}$

Q068 $0, \pi, 2\pi$

Q069 $\frac{\pi}{2}, \frac{3\pi}{2}$

Q070 $T=2\pi,\ -4\le y\le 4$

Q071 $T=\frac{2\pi}{5},\ -1\le y\le 1$

Q072 $T=4\pi,\ y\in\mathbb{R}$

Q073 horizontal compression by 1/2, reflection in x-axis and vertical scale 1/2; $T=\pi$

Q074 left by $\frac{\pi}{6}$, $T=2\pi$

Q075 right by $\frac{\pi}{2}$, $T=2\pi$

Q076 $\theta=-\frac{\pi}{4}$; $T=\pi$

Q077 right by $\frac{\pi}{4}$, $T=\frac{\pi}{2}$, $-2\le y\le 2$

Q078 left by $\frac{\pi}{2}$, $T=6\pi$

Q079 left by $\frac{\pi}{2}$, $T=\frac{\pi}{2}$

Q080 right by $\frac{\pi}{4}$, $T=4\pi$, $-2\le y\le 2$

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 right by $\frac{\pi}{2},\ T = \pi,\ -3 \le y \le 3$

Quiz WRITTEN 6 $\cos\theta - \sin\theta$

Quiz WRITTEN 7 $T = \pi;\ \theta = \frac{5\pi}{6} + k\pi$

Quiz WRITTEN 8 $T = \frac{\pi}{3},\ -2 \le y \le 0;$ reflected below the